

Statement of Work
AH1 Gas Fired Air Heater
16 APRIL 2026

Scope: To obtain a gas fired air heater to meet AFRL research needs of 2.5-34 lbm/sec air at a pressure range of 300-750 PSIG to be heated from 100 F up to a range of 250 – 1250 F. Temperature rating to be within minus zero and plus 4 percent tolerance. Vendor shall deliver a air heater to meet the requirement, provide commissioning and training (after installation into the flow system by the government), and provide O&M manuals along with a 1-year warranty after acceptance.

Background: Currently RQ uses high pressure, high temperature air delivery system to simulate altitude and velocity of propulsion systems designed for Air Force platforms. The current gas fired heater has come to end of service life and needs replacement. The gas fired air heater will deliver the necessary air temperature to complete Air Force research mission.

Requirements Characteristics: Basic salient features to include:

1. Heater shall be provided as complete heater-burner package, including integral forced draft burner, coils, heater trim, refractory, insulation, controls, fuel train and accessory components.
2. Heater-burner package shall be provided fully assembled ready for ground mounting and connection to process air intake/discharge, plant compressed air, electrical, natural gas, vent, and control lines.
3. Equipment design and accessory installations shall permit accessibility for maintenance and service.
4. Heater shall comply with NFPA 85.
5. Heater shall be designed for working pressure of 500 psig and maximum operating pressure of 750 psig. The air heater shall be capable of continuously producing heated compressed air at conditions listed below. Heater operation shall be stable at all points of operation within the ranges of temperatures, pressures and flows listed below.
 - a. Output Process Air Inlet Temperature: 100 degrees F
 - b. Output Process Air Inlet Flow Range: 2.5-34 lbm/sec
 - c. Maximum Process Air Discharge Temperature Range: 250 - 1250 degrees F
 - d. Maximum Heater Operation Period: up to 16 hours continuously.
6. The air heater shall be provided capable of operating daily and also capable of operating after a period of inactivity of up to 30 days with no damage to the heat exchanger or any other components.
7. Air heater shall be suitable for outdoor installation. Provide adequate weather and corrosion protection to ensure reliable service life over a 25 -year period, with minimal degradation due to exposure to outdoor ambient conditions. The air heater will be installed near an active roadway which will expose all areas of the heater to possible salt spray.
8. Air heater shall be equipped to provide safe and stable operation while achieving performance indicated when operating at extreme outdoor temperatures encountered by

the installation. Review historical weather database and provide equipment that can operate at extreme outdoor temperatures recorded over past 30-year period.

Please consult spec section 23 80 20.10 10 GAS-FIRED AIR HEATERS for full requirements and features along with design condition assumptions.